Generative Adversial Networks (GAN)

FNN
RNN
CNN
Transformer
GAN

1000 dogs images

real data

GAN

fake data

looks like real data

→ training unstable
→ hard to evaluate

⇓

Diffusion models

i) rare dataset → training
ii) Ghibli trend
iii) Deep fakes

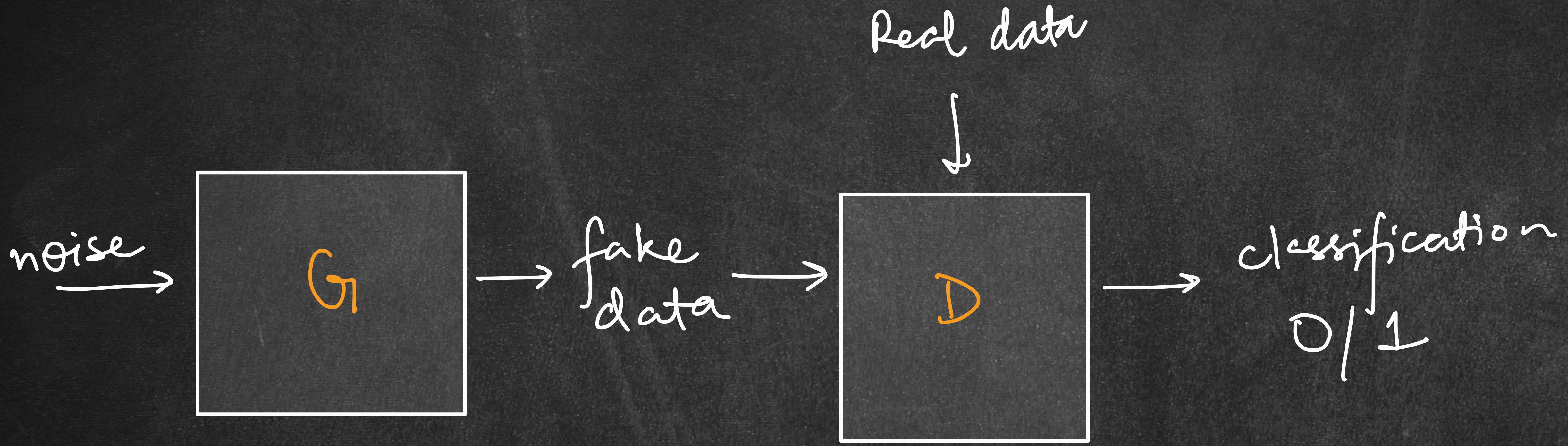
GANs

Architecture

2 NN → Generator (G)

2 NN → Discriminator (D)

GANs → Generative Adversial Networks



GANs

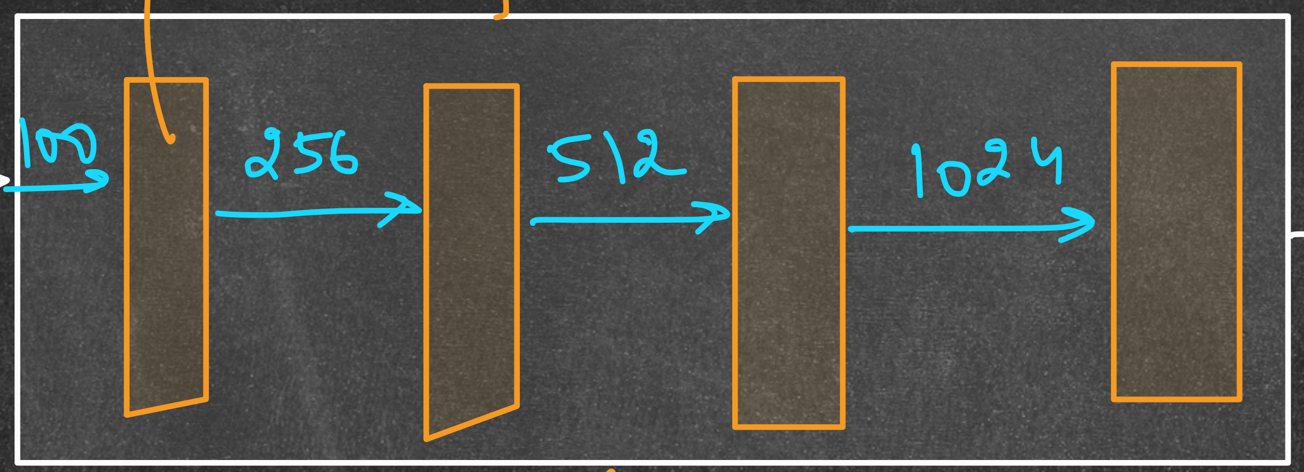
Generator Network → Vanilla GANs

noise → 

→ structured fake data

$y = Wz + b$

fully connected (dense)

noise → 

→ image (RGB)

64x64x3

↓

random nums

$[0.1, 0.32, 0.9, \ldots] = z$

100

G

↓

UPSAMPLING (expansion)

gaussian ($\mu = 0$, std dev $= 1$)

uniform

⋮

GANs

Generator Network

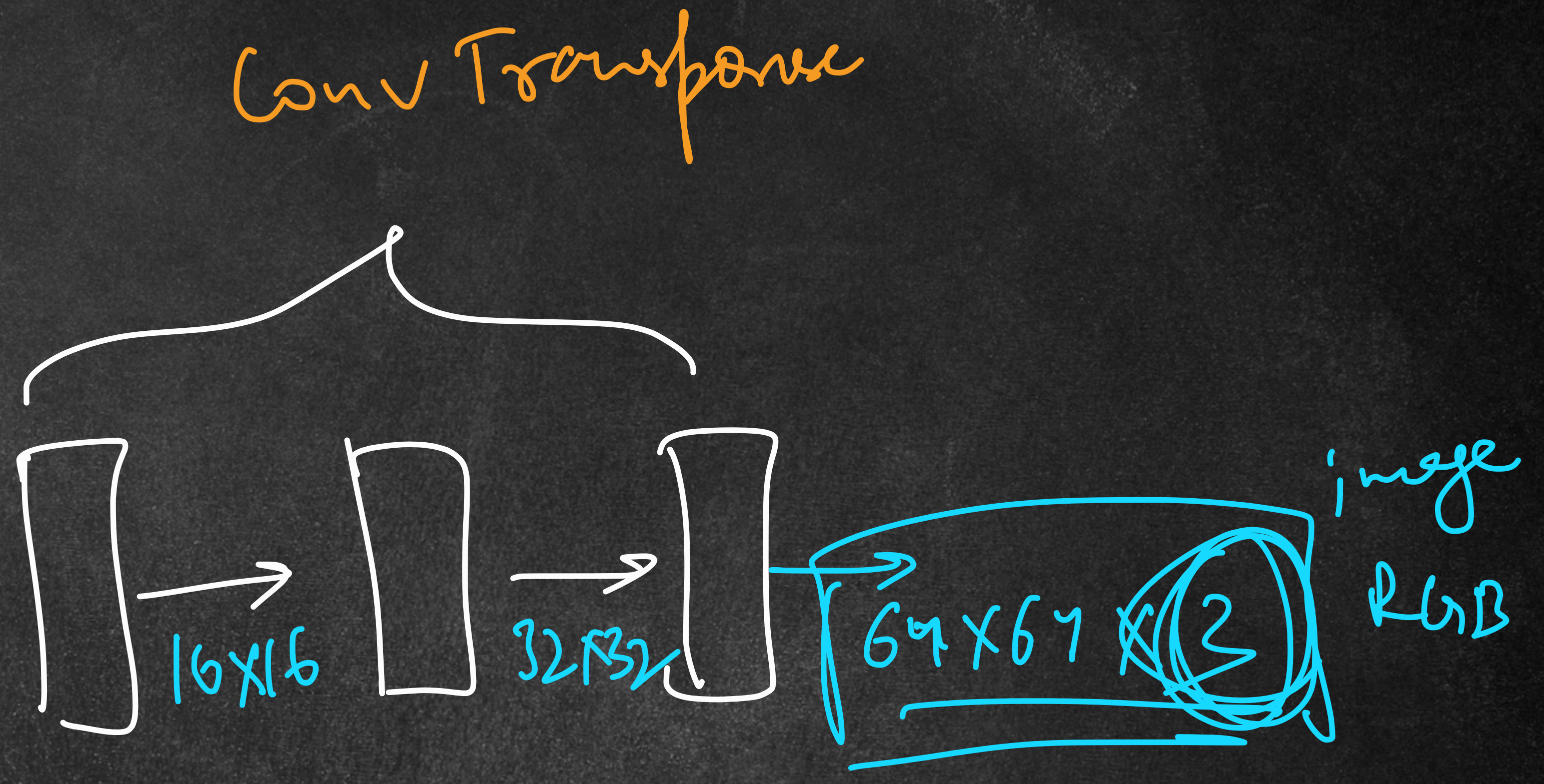
↓

DCGAN → Deep Convolutional GAN

noise (z) → (Dense) Fully connected → Reshape (Feature map) →

$X = Wz + b$

$8 \times 8 \times 128$



GANs → fake faces

APNA COLLEGE

Generator Network - **Code**

178x178

celeba images ⇒ 178 x 218

square

GANs

Discriminator Network

① Generator Loss (minimize)

$$g\_loss = -\log(D(G(z)))$$

① $\approx \infty$

② ≈ 0 ✓

② Discriminator Loss

$$d\_loss = -\left[\log(D(x)) + \log(1 - D(G(z)))\right]$$

$$-\left[\log(1) + \log(1-0)\right] \approx 0$$

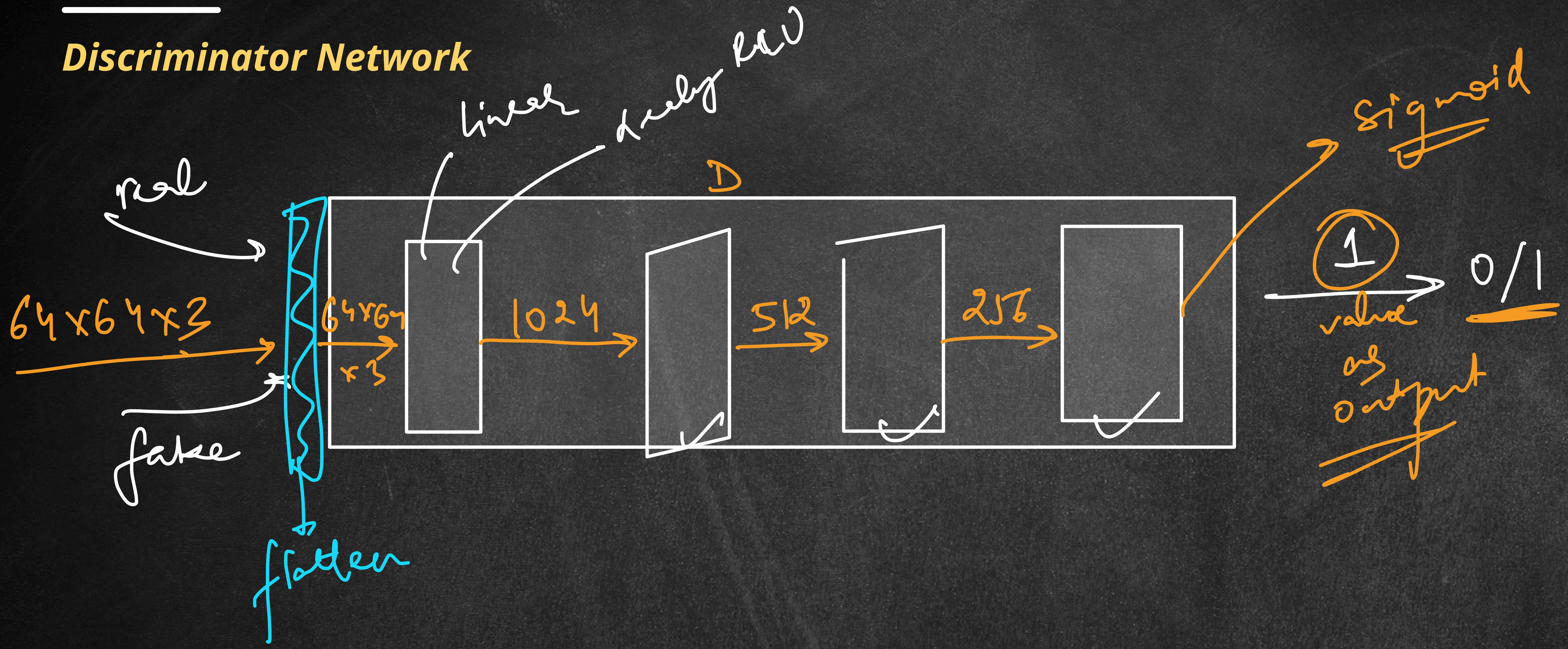
GANs

Discriminator Network

→ vanilla GAN

CNN

Binary Cross Entropy



G → upsampling → generate

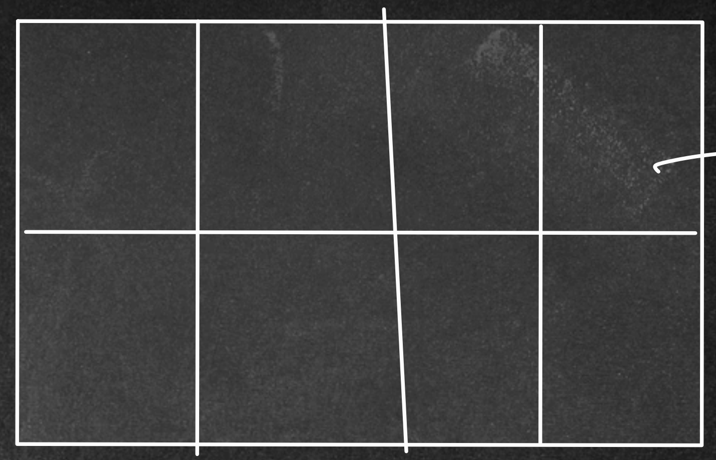
D → downsampling → classify

GANs

***Discriminator Network* - Code**

GANs

Training



$\rightarrow$ nrow = 4

grid tensor $\Rightarrow$ $\underset{0}{\text{channels}} \times \underset{1}{H} \times \underset{2}{W}$

matplotlib $\Rightarrow$ $\underset{1}{H} \times \underset{2}{W} \times \underset{0}{C}$